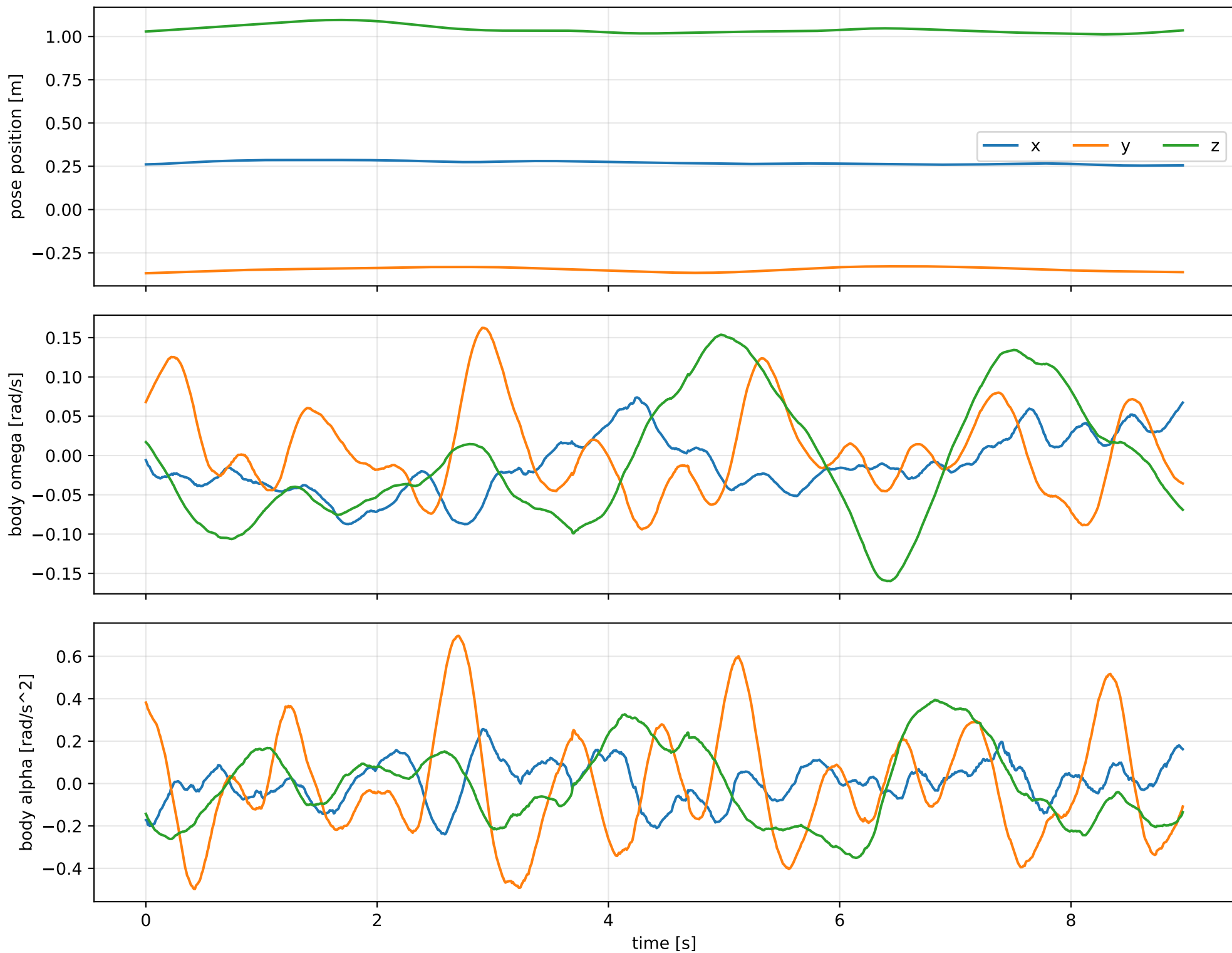
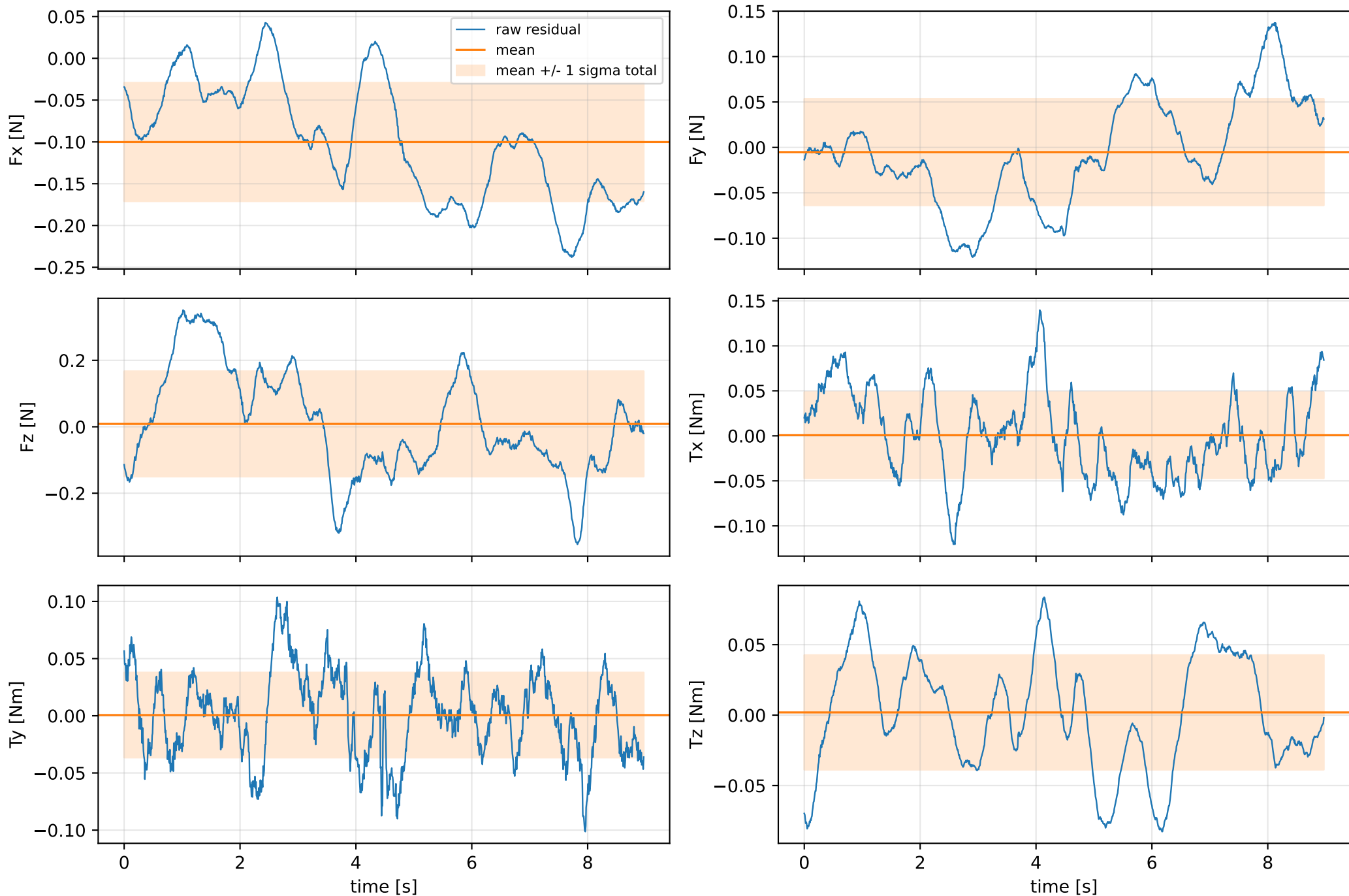


cog_prior_reseed_refined: SG trajectory and derived motion



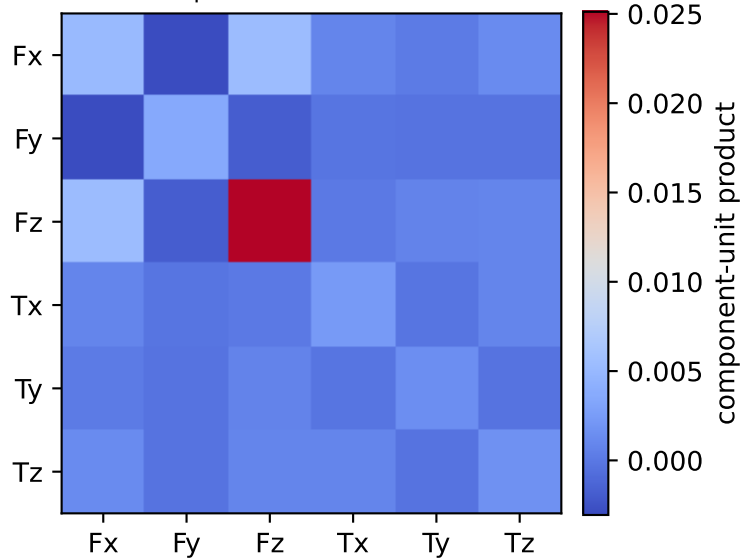
cog_prior_reseed_refined: raw Newton--Euler residual wrench (not trajectory-fitted external wrench)
mass gauge fixed to nominal mass = 2.35159759 kg



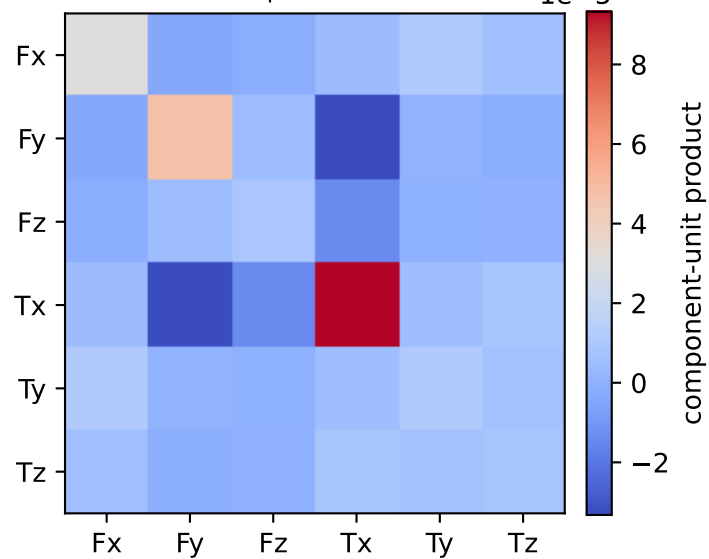
vector RMS about zero: force 0.20897 N, torque 0.072914 Nm; component std about mean: force [0.0709 0.0586 0.1585], torque [0.0478 0.0372 0.0406]

cog_prior_reseed_refined: residual-wrench covariance decomposition in nominal-mass gauge
force [N], torque [Nm]; raw excess eigenvalues [0.001 0.001 0.001 0.003 0.006 0.027]

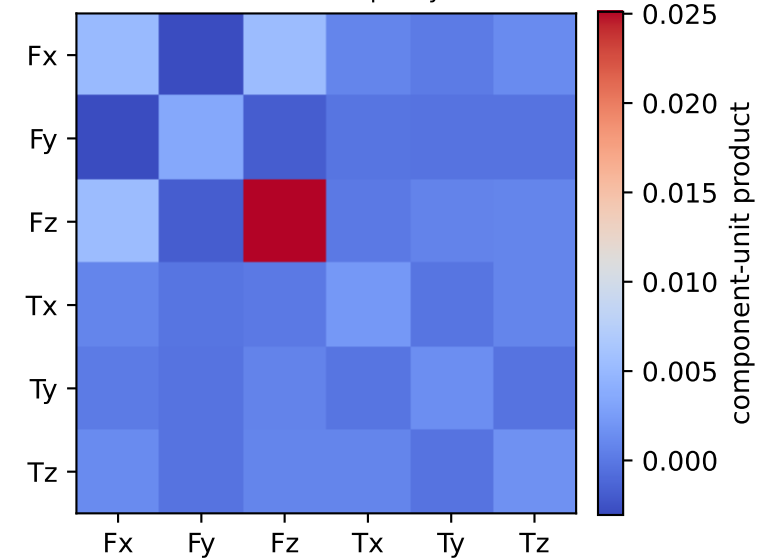
empirical total covariance



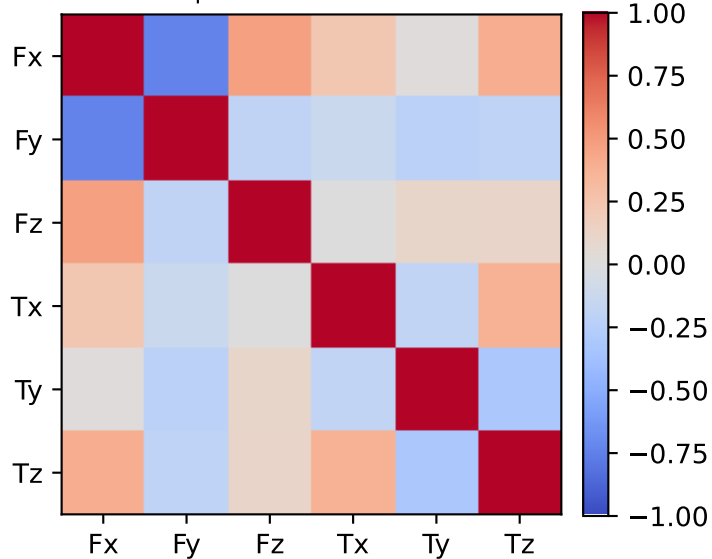
mean full-SG prediction covariance $1e-5$



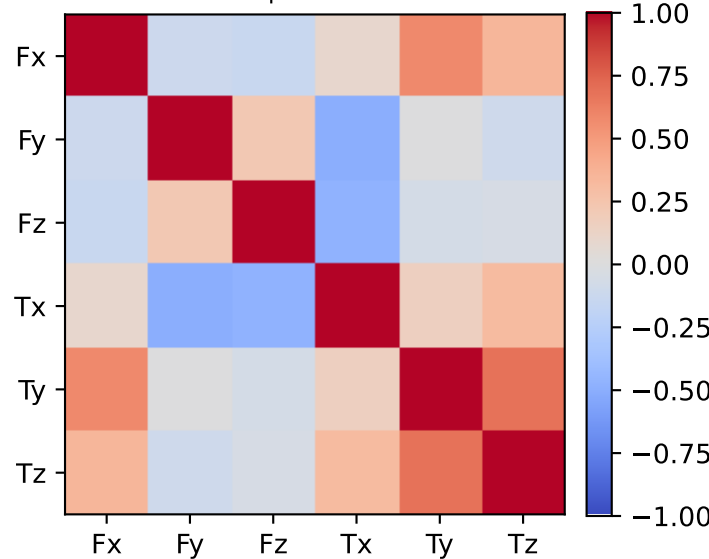
PSD excess model discrepancy covariance



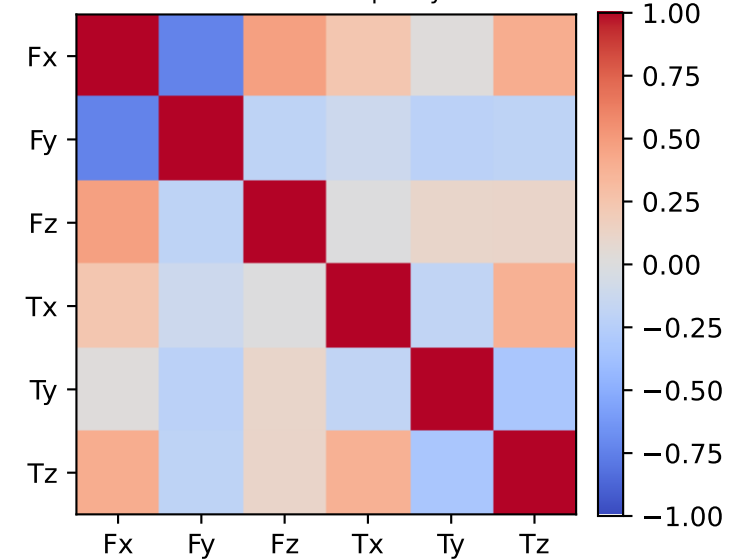
empirical total correlation



mean full-SG prediction correlation

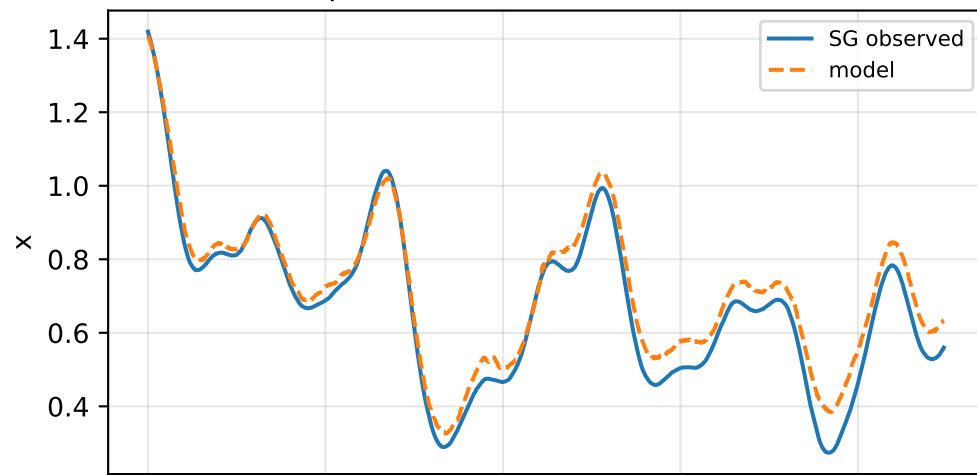


PSD excess model discrepancy correlation

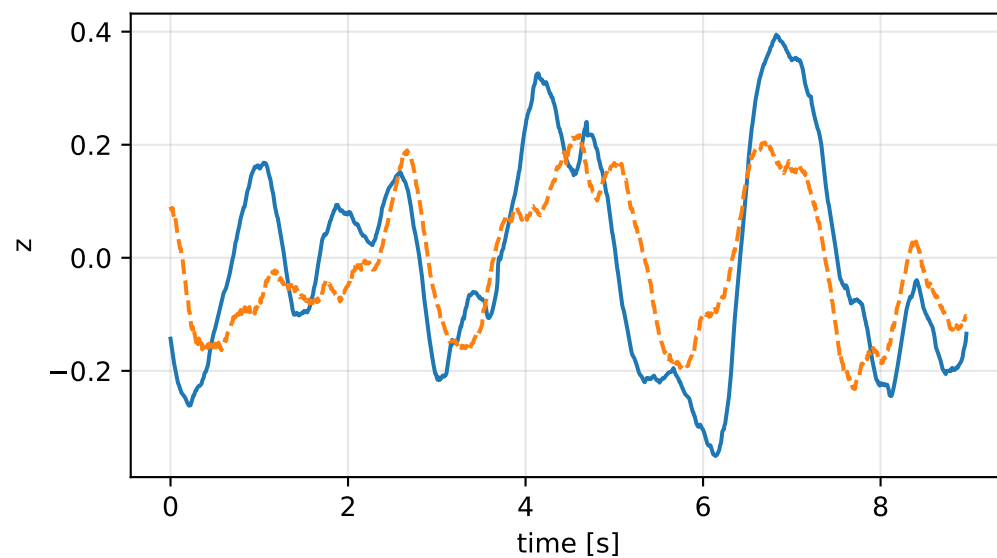
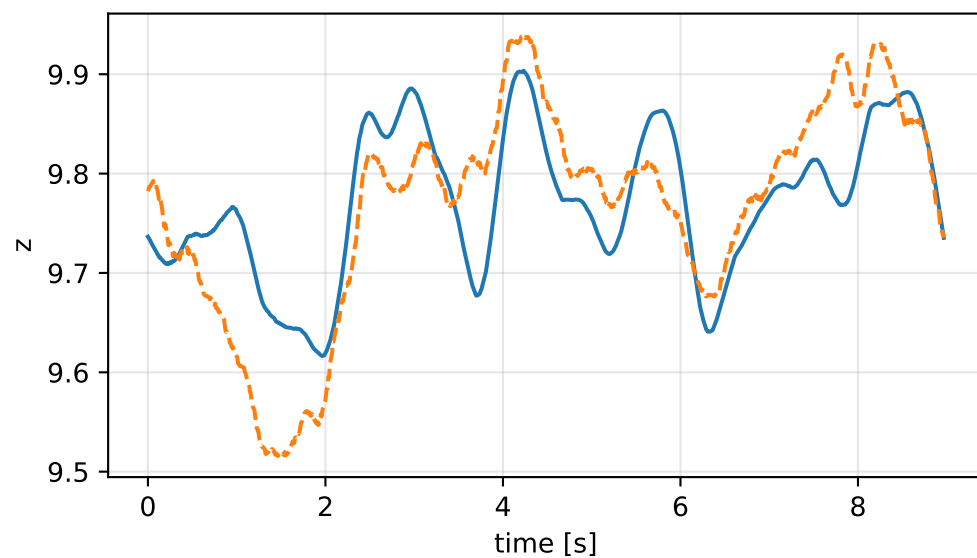
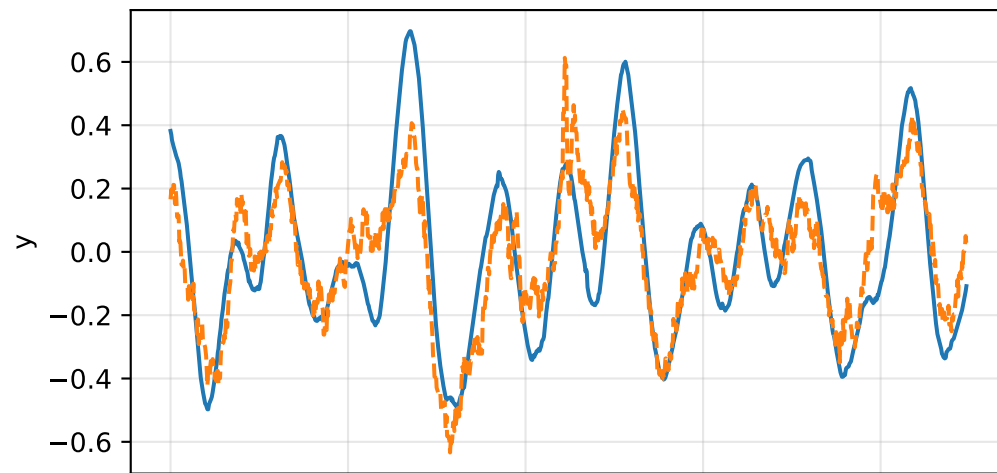
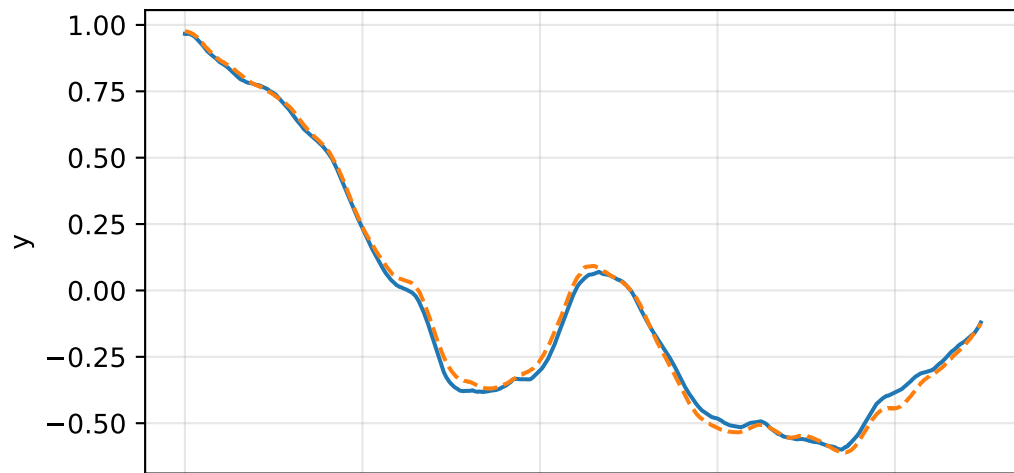
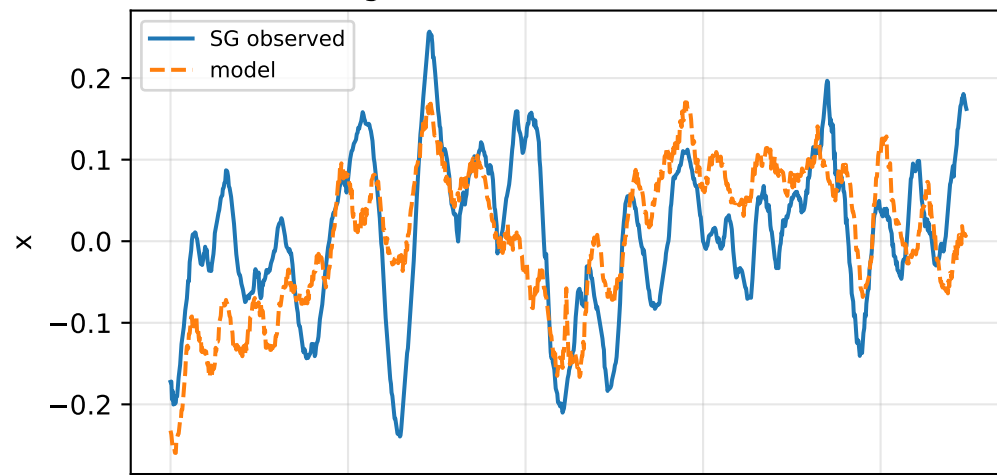


cog_prior_reseed_refined: acceleration residual objective

specific acceleration [m/s²]

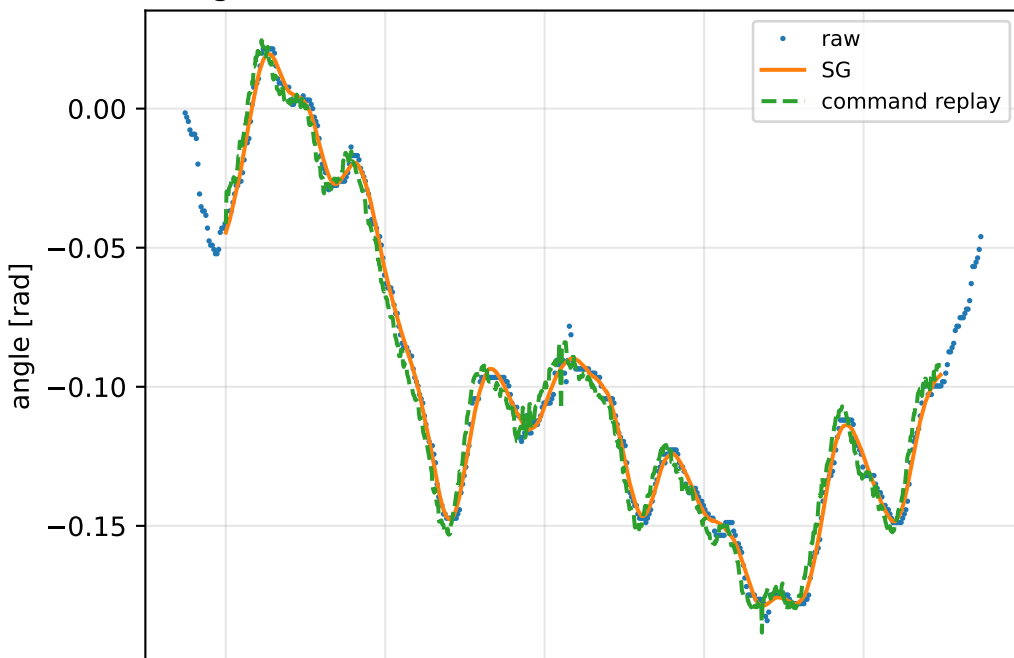


angular acceleration [rad/s²]

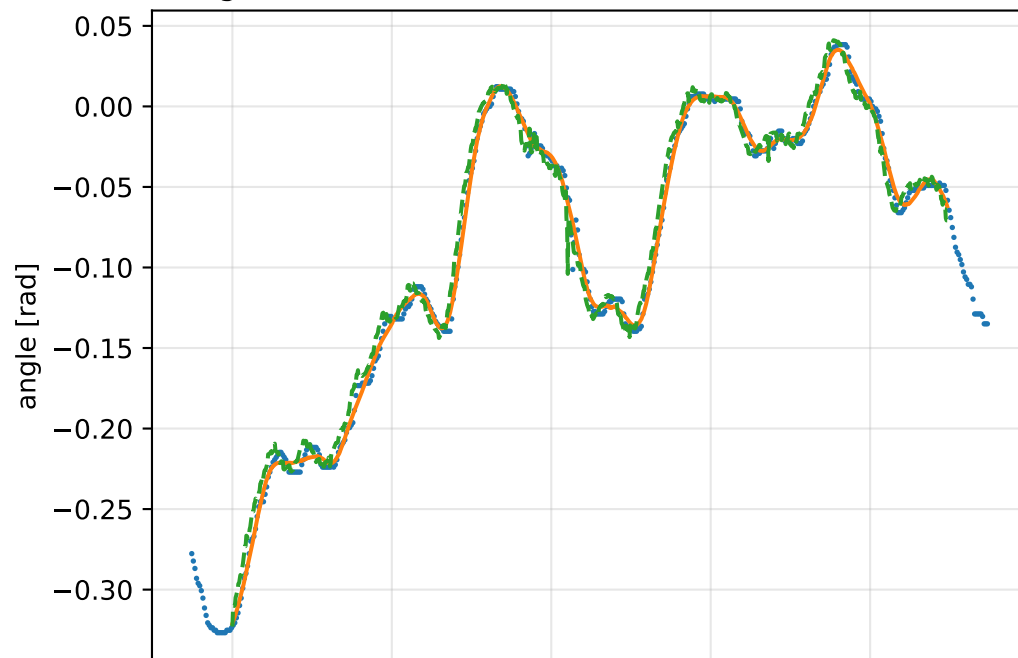


cog_prior_reseed_refined: gimbal measurement audit (objective=measured_sg)

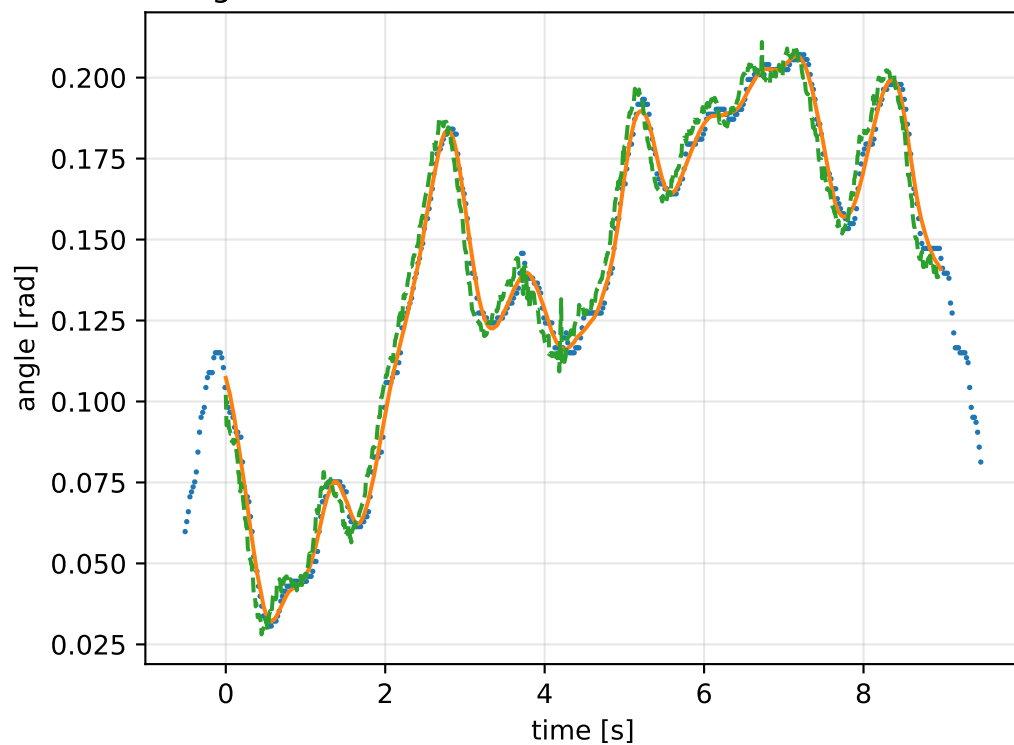
gimbal 1: RMSE=0.006345, max=0.01494 rad



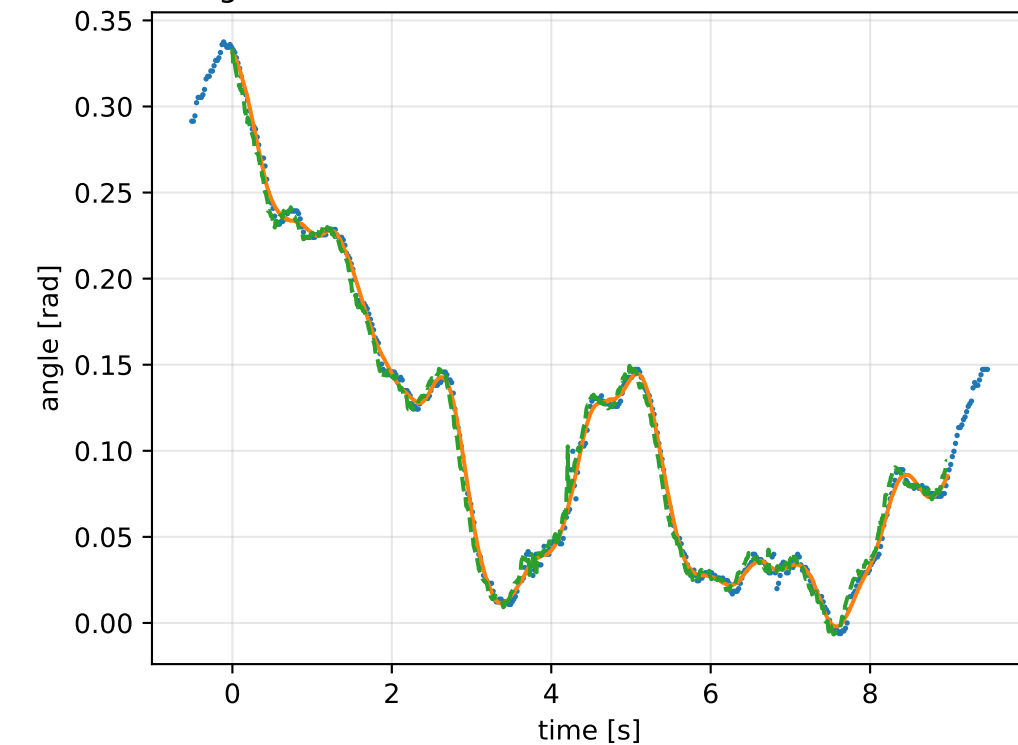
gimbal 2: RMSE=0.008835, max=0.04462 rad



gimbal 3: RMSE=0.007019, max=0.01611 rad

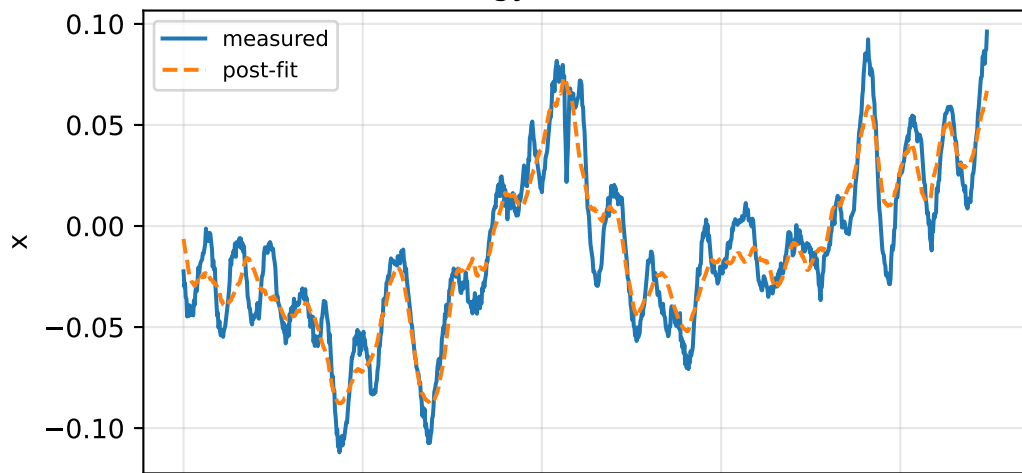


gimbal 4: RMSE=0.006321, max=0.03642 rad

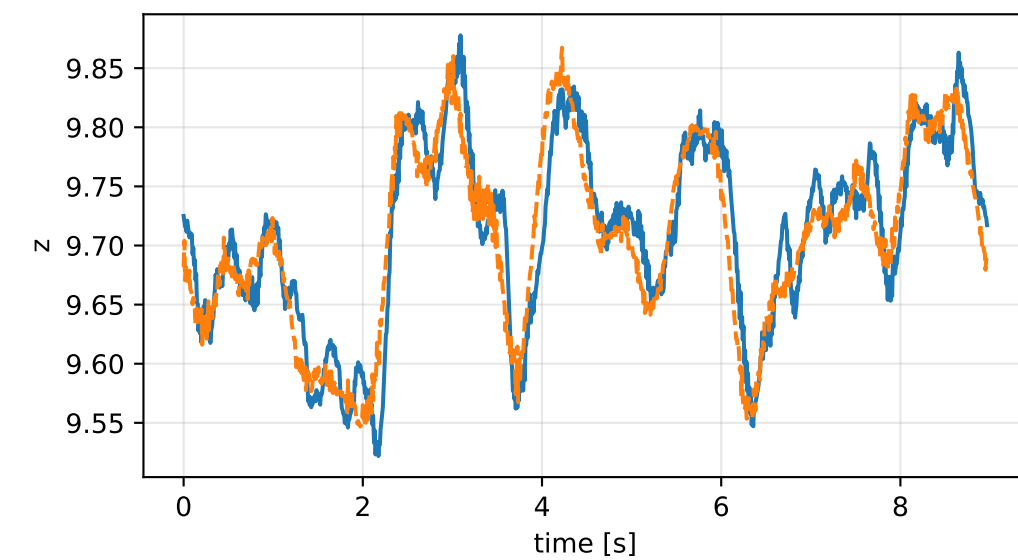
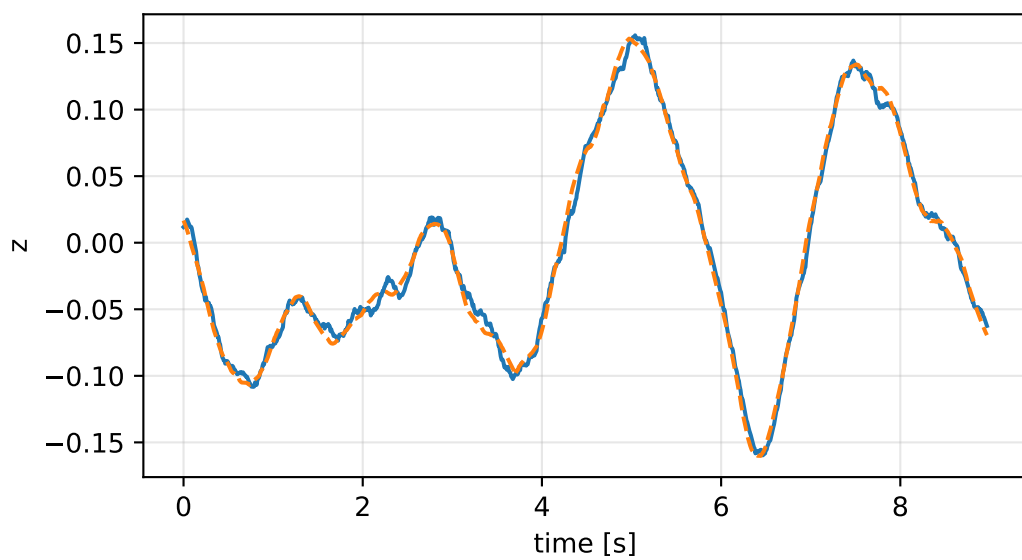
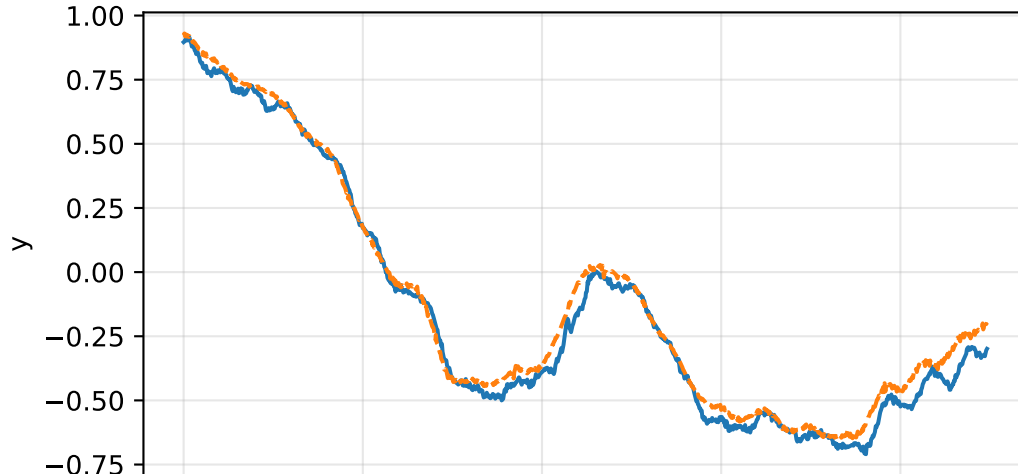
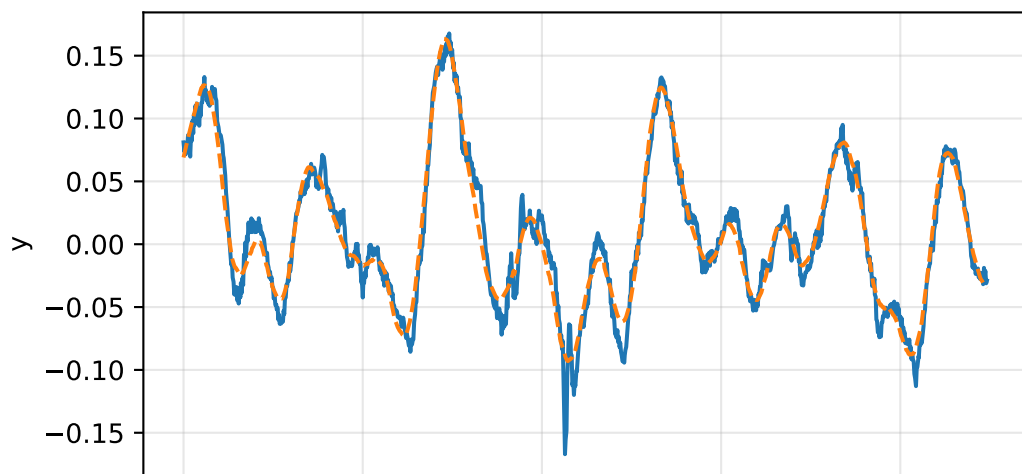
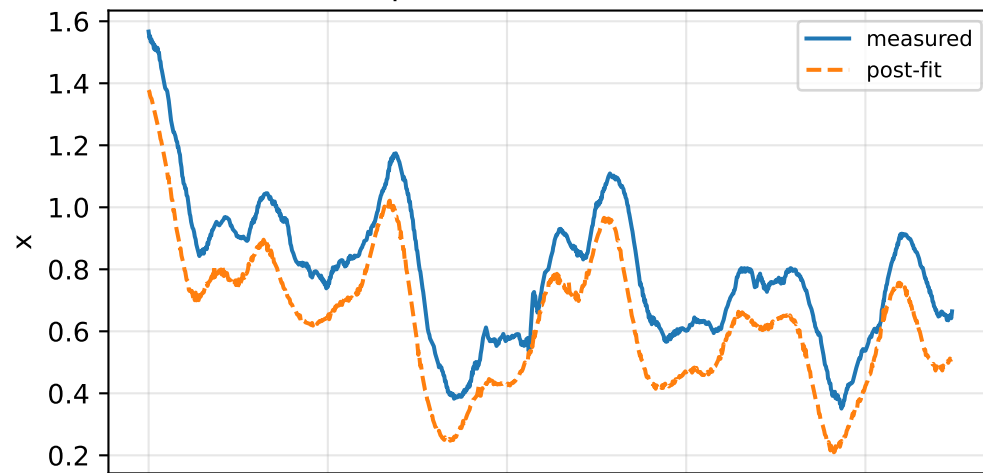


cog_prior_reseed_refined: IMU comparison (diagnostic only)

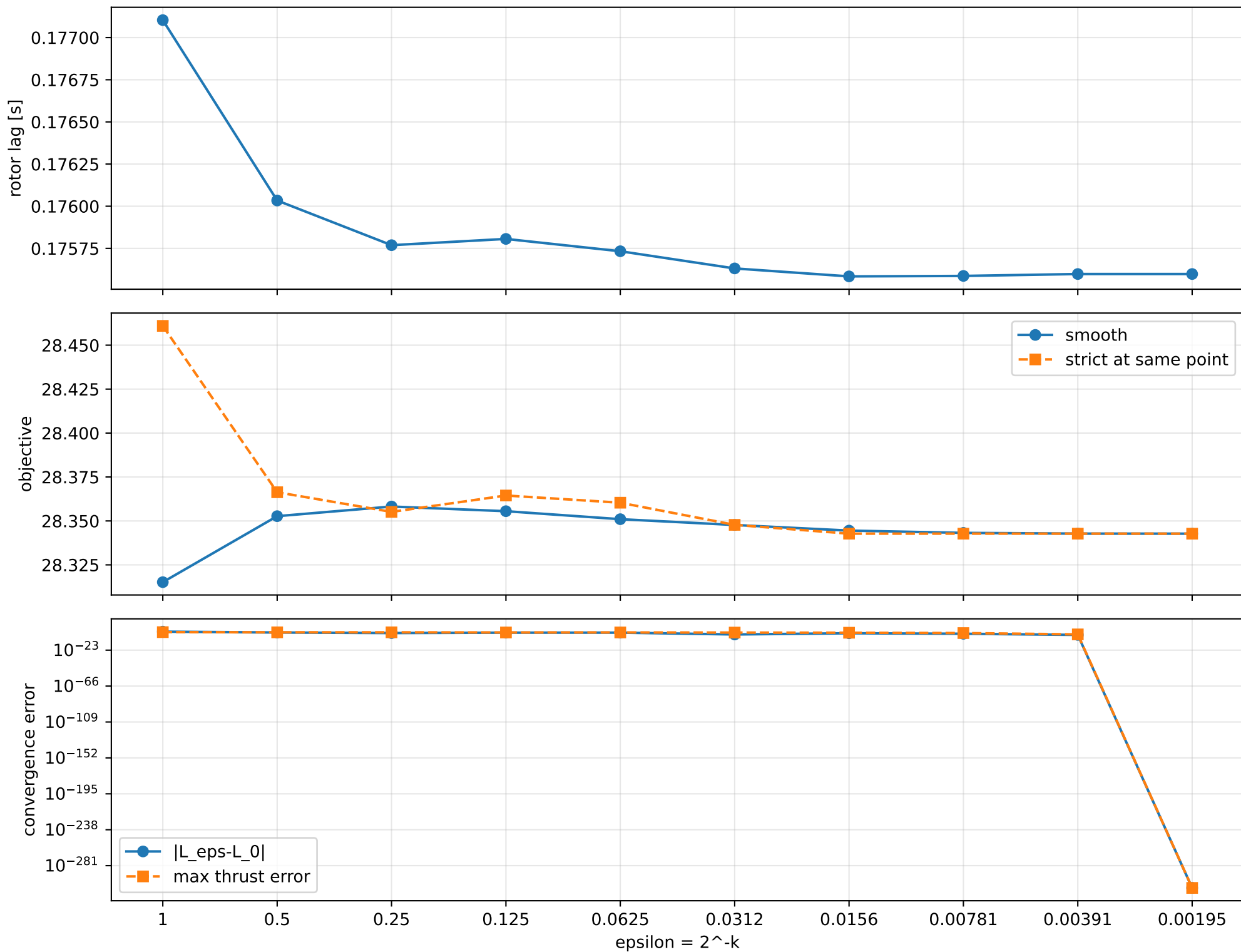
gyro [rad/s]



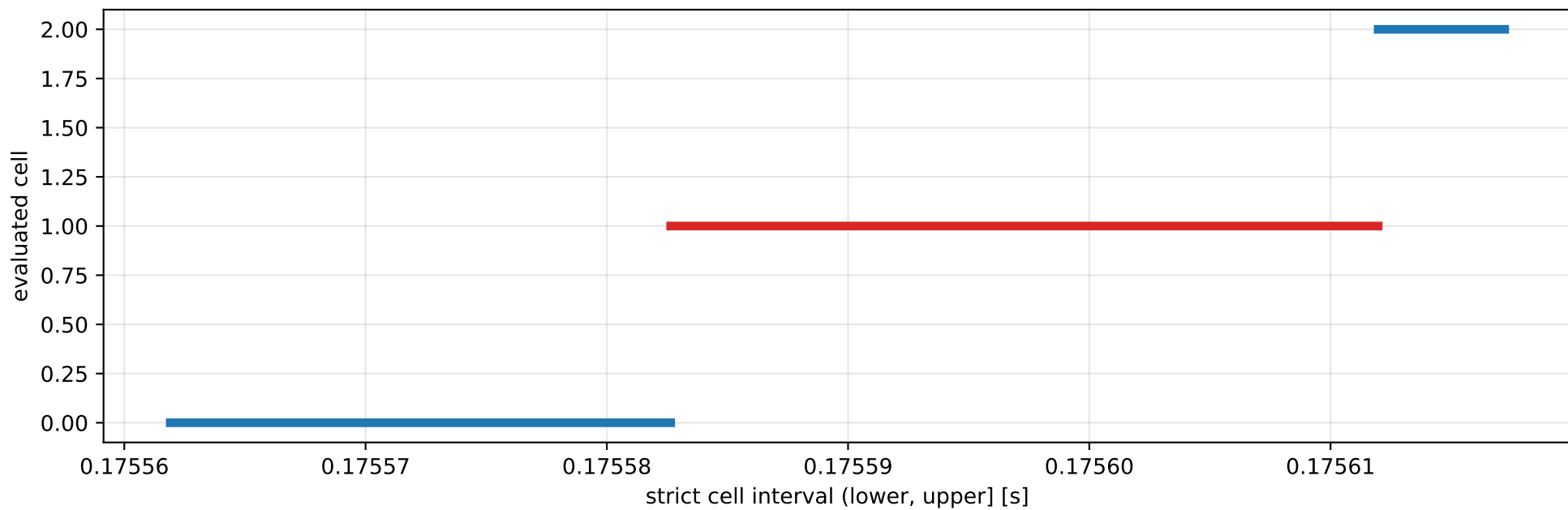
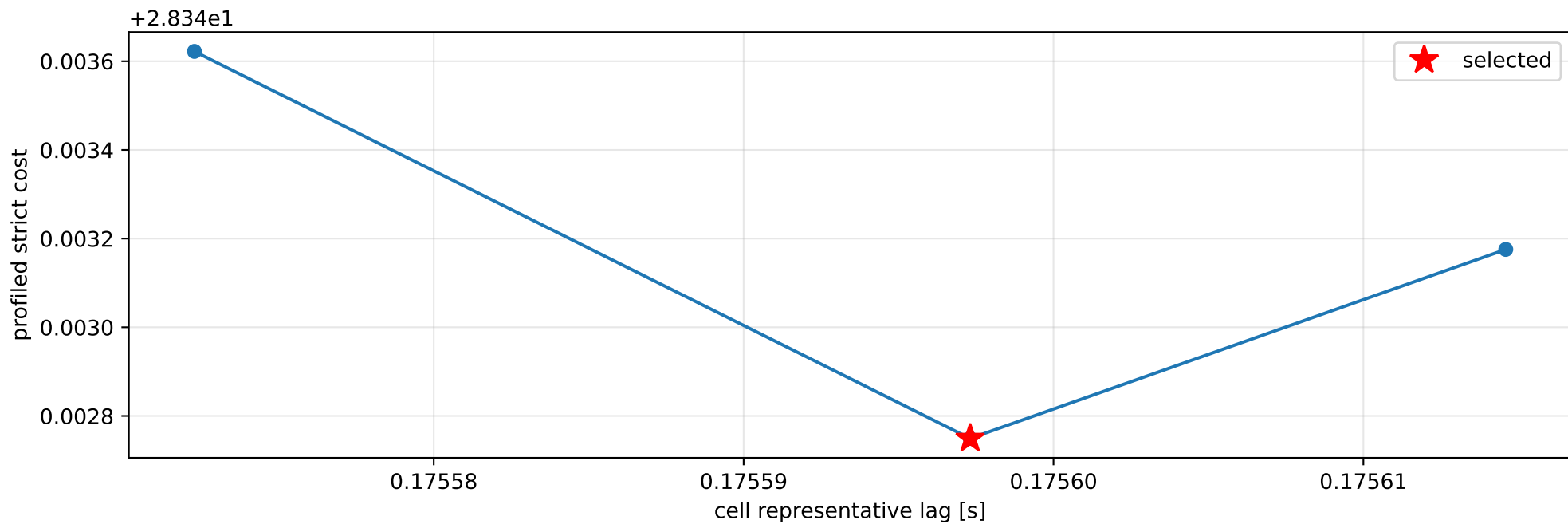
specific force [m/s^2]



cog_prior_reseed_refined: smooth-to-strict continuation



cog_prior_reseed_refined: exact strict-ZOH lag cells



selected (0.175582647, 0.175611973] s; final smooth 0.175597984 s; data boundary=False

cog_prior_reseed_refined: parameter estimate

Case: cog_prior_reseed_refined

status: completed (all stages completed)

mass [kg]: nominal 2.35159759; estimated 6.21658829

inertia [kg m²]:

```
[[ 1.44194598 -0.03158833  0.11197266]
 [-0.03158833  0.68175385 -0.01894932]
 [ 0.11197266 -0.01894932  0.80121865]]
```

CoG body [m]: [-0.01981537 -0.05513406 -0.00502729]

force effectiveness: [2.76340042 2.70438557 1.42675263 1.73237303]

scale-free J/m [m²]:

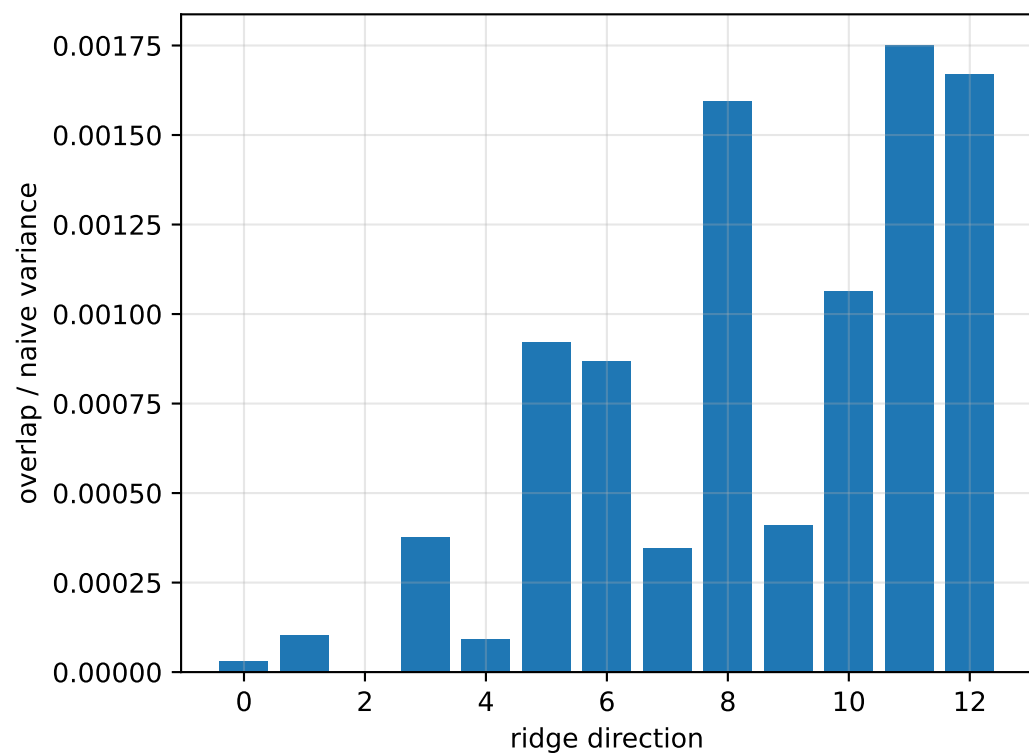
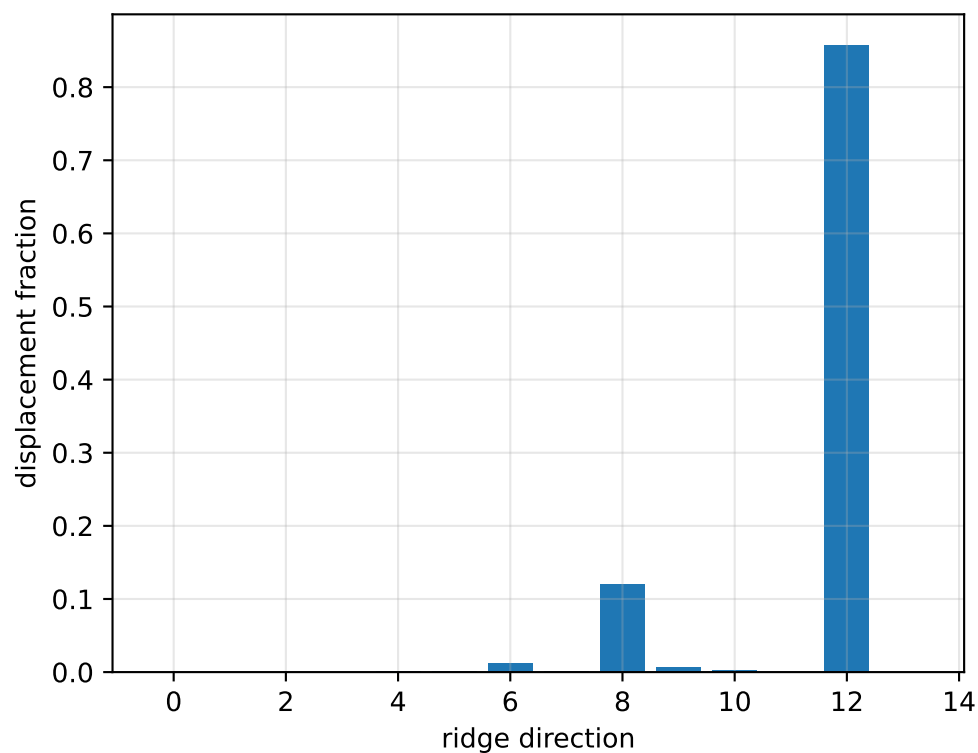
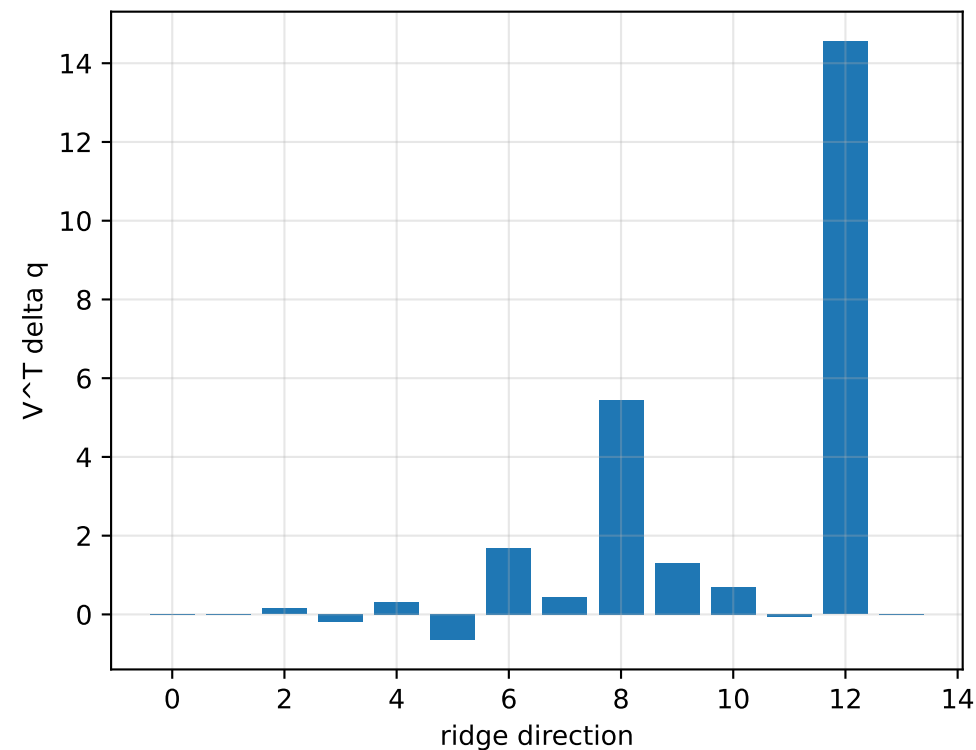
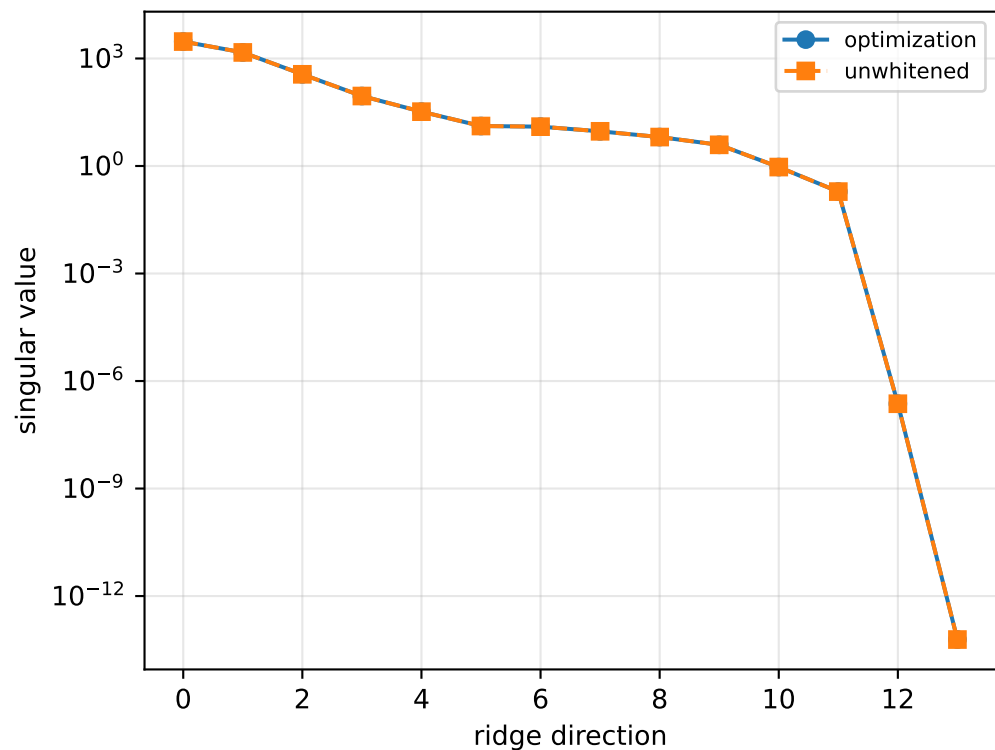
```
[[ 0.23195134 -0.0050813  0.01801192]
 [-0.0050813  0.10966688 -0.00304819]
 [ 0.01801192 -0.00304819  0.12888398]]
```

scale-free f/m: [0.44452042 0.43502729 0.22950734 0.27866942]

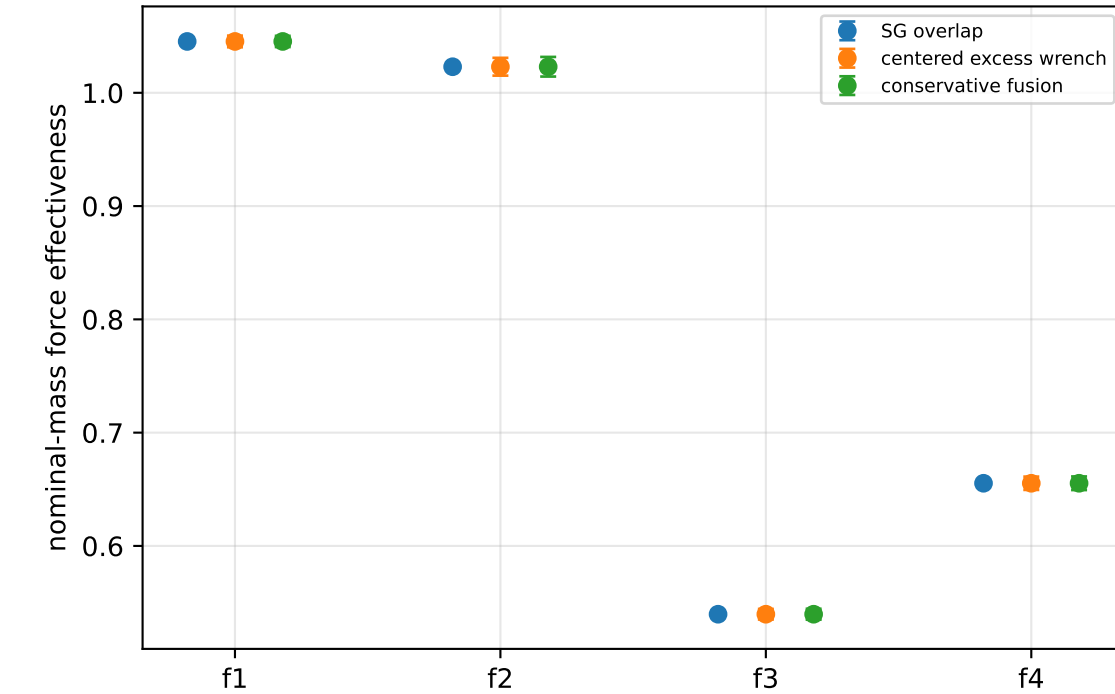
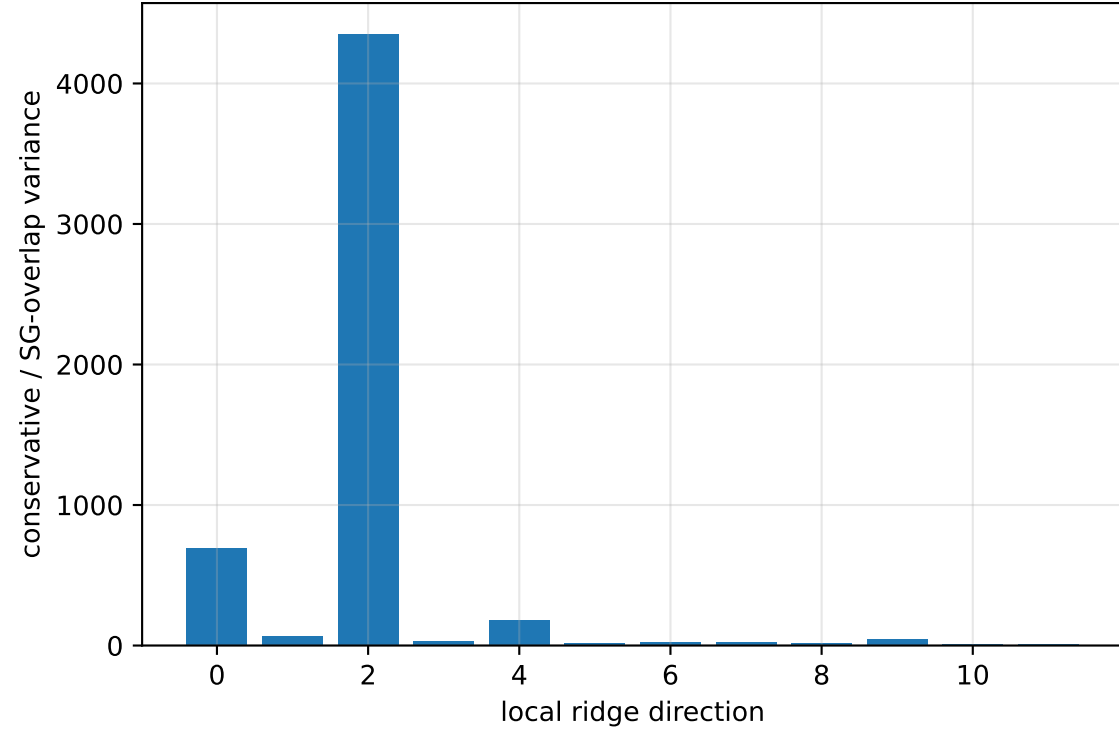
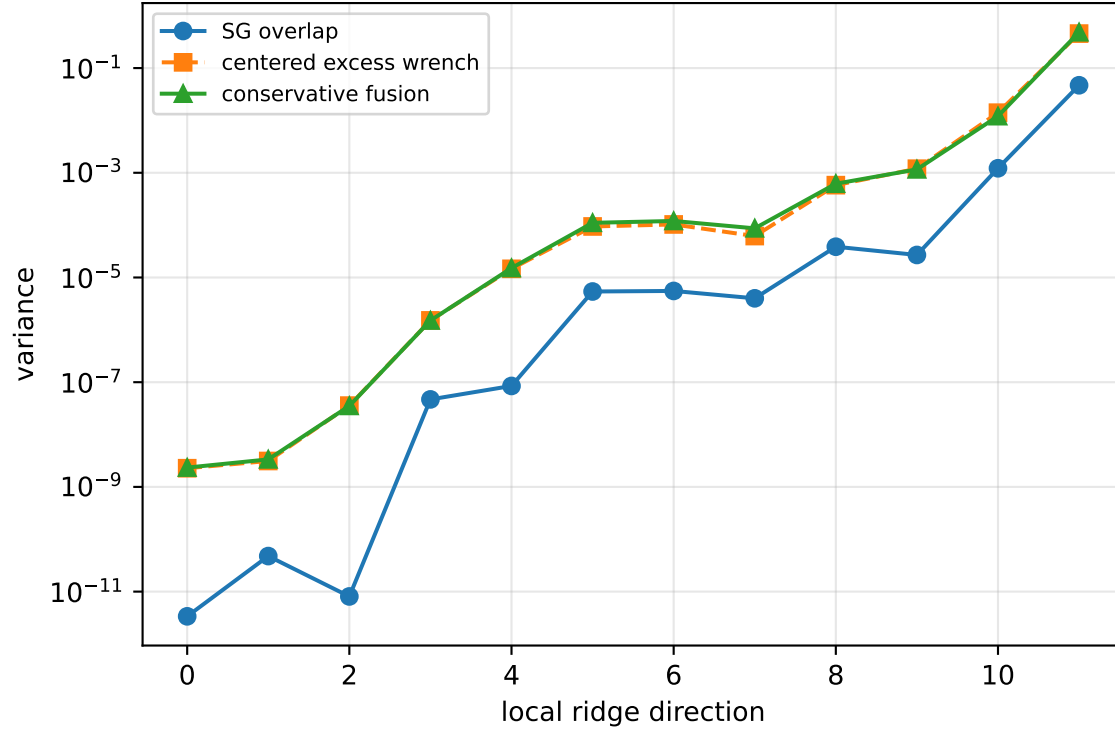
rotor lag cell: (0.175582647, 0.175611973] s

representative: 0.17559731 s

cog_prior_reseed_refined: ridge spectrum (rank 13, nullity 1, ||jv||=7.68e-14)



cog_prior_reseed_refined: Conservative fusion uncertainty



Top three non-gauge ambiguous directions
 $\text{tr}(\mathbf{M}_{\text{res}}) / \text{tr}(\mathbf{M}_{\text{SG}}) = 401.6$
wrench-to-acceleration recovery max error = $2.75\text{e-}15$
exact scale gauge ridge index = 13 (alignment 1)

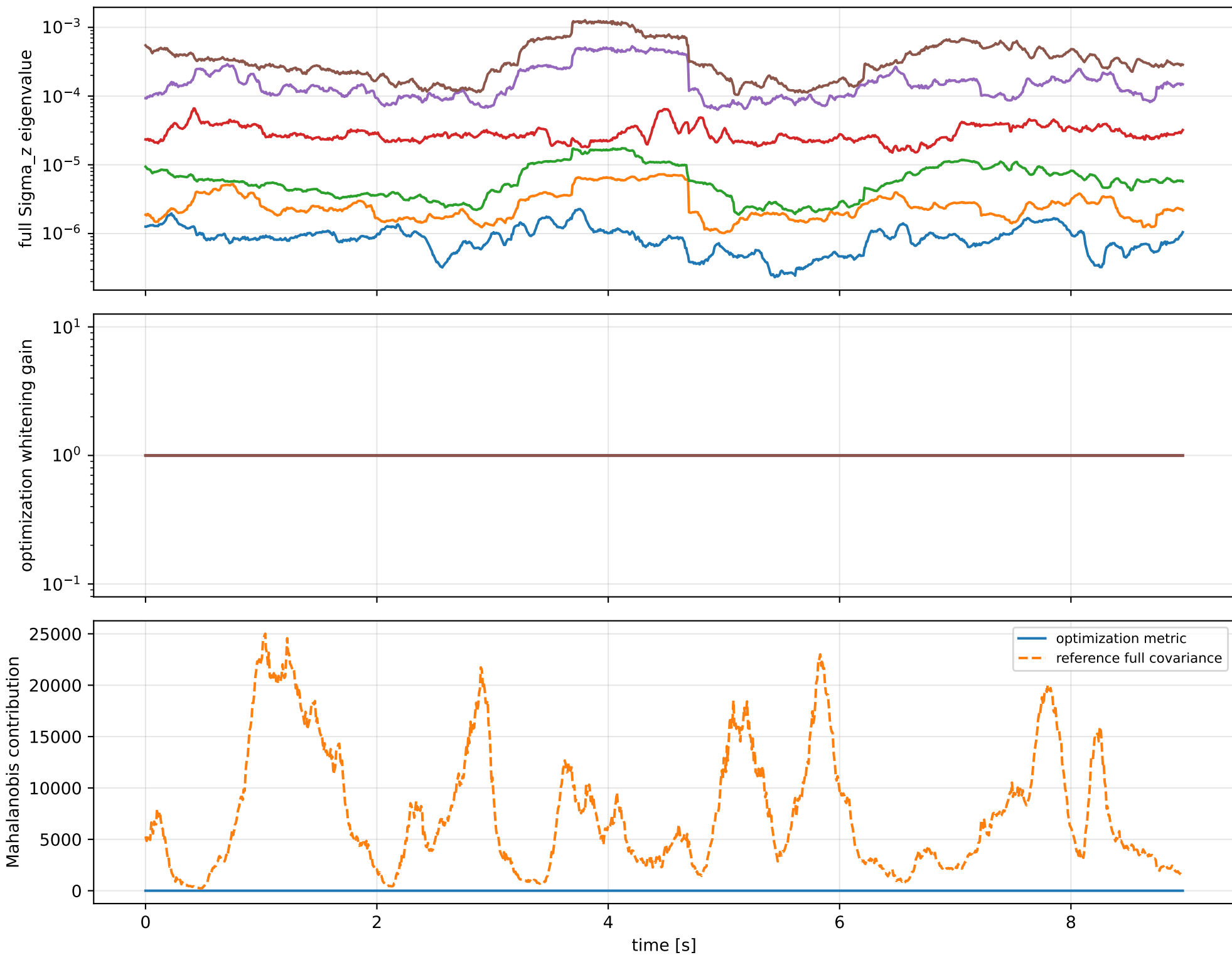
index 11: variance=0.4828, ratio=10.29
 $\log_{\text{mass}}=+0.00153$; second_moment_diag_1=+0.0645
second_moment_diag_2=-0.0601; second_moment_diag_3=-0.0125
second_moment_offdiag_12=+0.992; second_moment_offdiag_13=-0.0877
second_moment_offdiag_23=+0.0234; cog_x=-0.000544
cog_y=+0.000263; cog_z=-0.000366
log_force_eff_1=+0.00324; log_force_eff_2=-0.00238
log_force_eff_3=+0.00146; log_force_eff_4=+0.00428

index 10: variance=0.01206, ratio=9.893
 $\log_{\text{mass}}=-0.0182$; second_moment_diag_1=-0.0145
second_moment_diag_2=-0.0435; second_moment_diag_3=+0.139
second_moment_offdiag_12=+0.0883; second_moment_offdiag_13=+0.982
second_moment_offdiag_23=-0.0422; cog_x=+0.00573
cog_y=-0.000785; cog_z=-0.000616
log_force_eff_1=-0.0343; log_force_eff_2=+0.00554
log_force_eff_3=+0.0192; log_force_eff_4=-0.0537

index 9: variance=0.001161, ratio=43.1
 $\log_{\text{mass}}=-0.169$; second_moment_diag_1=+0.001
second_moment_diag_2=+0.893; second_moment_diag_3=-0.0379
second_moment_offdiag_12=+0.0532; second_moment_offdiag_13=+0.038
second_moment_offdiag_23=+0.216; cog_x=-0.00429
cog_y=-0.000447; cog_z=-0.0111
log_force_eff_1=-0.163; log_force_eff_2=-0.172
log_force_eff_3=-0.212; log_force_eff_4=-0.14

The conservative fusion covariance deliberately retains the existing SG-overlap uncertainty and adds the uncentered empirical residual score second moment without subtracting the SG contribution. It is intended as a conservative fusion distribution, not as a calibrated generative noise covariance.

cog_prior_reseed_refined: optimization vs reference covariance



cog_prior_reseed_refined: wrench / closure (force max $1.71\text{e-}14$, torque max $9.71\text{e-}17$)

